

Cloud Application Development – Security Lab

Securing a Cloud Virtual Machine (VM)

Lab Objective

The objective of this lab is to understand and implement **basic security best practices** when deploying applications on cloud virtual machines. Students will learn how to **secure access, networking, operating system, and applications** running on a cloud VM.

Learning Outcomes

After completing this lab, students will be able to:

- Secure access to cloud VMs using SSH keys
- Configure firewall and security group rules
- Apply OS-level hardening techniques
- Implement basic application security measures
- Monitor and audit VM activity

Cloud Platform

You may use **any one** of the following:

- AWS EC2
- Azure Virtual Machine
- Google Compute Engine
- OR a **local private cloud** (Proxmox / OpenStack) for teaching purposes

Lab Scenario

You are a **cloud engineer** deploying a **web application** on a cloud VM. Your task is to ensure the VM and the application are **secure against common attacks**.

Part 1: Secure VM Provisioning

Step 1: Create a Cloud VM

- OS: **Ubuntu Server 22.04**
- VM Size: Small (1 vCPU, 1–2 GB RAM)
- Public IP enabled

 **Security Concept:** Minimal resources reduce attack surface.

Part 2: Secure Access to the VM

Step 2: Use SSH Key-Based Authentication

- Generate SSH key pair
- Disable password-based login

```
sudo nano /etc/ssh/sshd_config
```

Set:

```
PasswordAuthentication no  
PermitRootLogin no
```

Restart SSH:

```
sudo systemctl restart ssh
```

 **Why?**

Prevents brute-force and password-guessing attacks.

Video Links

- <https://www.youtube.com/watch?v=bYrIL57NUZU>
- <https://www.youtube.com/watch?v=eFA7Kd8Oqok>

Part 3: Network Security (Firewall & Security Groups)

Step 3: Configure Cloud Security Group

Allow only:

- SSH → Port 22 (your IP only)
- HTTP → Port 80
- HTTPS → Port 443

❌ Block all other ports

🔒 Why?

Limits network exposure to only required services.

Step 4: Configure OS Firewall (UFW)

```
sudo ufw allow 22
sudo ufw allow 80
sudo ufw allow 443
sudo ufw enable
sudo ufw status
```

🔒 Defense in Depth: Cloud firewall + OS firewall.

Part 4: Operating System Hardening

Step 5: Update and Patch the OS

```
sudo apt update && sudo apt upgrade -y
```

Enable automatic updates:

```
sudo apt install unattended-upgrades
```

🔒 Why?

Prevents exploitation of known vulnerabilities.

Step 6: Create a Non-Root User

```
sudo adduser clouduser
sudo usermod -aG sudo clouduser
```

🔒 Principle of Least Privilege

Part 5: Application-Level Security

Step 7: Deploy a Simple Web Application

Example: Nginx

```
sudo apt install nginx -y
```

Verify access via browser.

Step 8: Enable HTTPS (SSL/TLS)

- Use Let's Encrypt

```
sudo apt install certbot python3-certbot-nginx  
sudo certbot --nginx
```

 Why?

Encrypts data in transit and prevents MITM attacks.

Part 6: Protection Against Common Attacks

Step 9: Install Fail2Ban

```
sudo apt install fail2ban -y  
sudo systemctl enable fail2ban
```

 Why?

Blocks repeated failed login attempts.

Step 10: Disable Unused Services

```
sudo systemctl list-unit-files --type=service
```

Disable unnecessary services:

```
sudo systemctl disable service_name
```

Part 7: Logging and Monitoring

Step 11: Enable Logging

- Check authentication logs:

```
sudo tail -f /var/log/auth.log
```

- Monitor system activity:

```
top
```

 Why?

Security is incomplete without monitoring.

Part 8: Backup and Recovery

Step 12: VM Snapshots

- Create VM snapshot (cloud console)
- Restore VM from snapshot (simulation)

 Why?

Protects against ransomware and accidental data loss.

Video Links

- <https://youtu.be/QDxRRpelA0c?si=miorwwcy3OYamT1U>
- <https://youtu.be/t97LC8UKT0o?si=HdHuHByB4Rmp0Gjp>

Lab Questions (Assessment)

1. Why is SSH key-based authentication more secure than passwords?
2. Explain the concept of **Defense in Depth** in this lab.
3. What risks exist if all ports are left open?
4. How does HTTPS improve application security?
5. Why is monitoring important in cloud security?